

Modular Forms

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September 1, 2026

Preface

Today, September 1st 2026, marks my 25th birthday. Usually, resolutions are made around New Year's, but to me this seemed like an appropriate time as well. I have self-studied (pure) mathematics on the side before, but never thoroughly nor was it well-structured. This time I intend to change that and aim to study the lecture notes on modular forms by Dr P. Bruin and Dr S. Dahmen entirely and complete all the problems. In this document I will write my solutions and include thoughts I have whilst reading the text.

I am in no rush, so an additional challenge is that I will not use any form of AI in solving the problems. The reason is simply that I notice that it becomes too easy and I believe in the value of the struggle of not knowing how to tackle a problem. Of course, I will allow myself to use the more 'classic' sources such as Wikipedia, StackExchange and papers. For this reason, my solutions might contain errors.¹ This project really is a challenge to me and my patience. Knowing myself, I would love to learn all sorts of topics in pure mathematics, so preferably as fast as possible too. To that end, I force myself to only focus on Modular Forms and not use AI to solve problems quicker. I believe (or hope) that the result will be more rewarding than superficially studying several subjects at once in a short time. After all, this project is my own so who am I kidding in that case?

Anyhow, I'm rambling a bit. Why modular forms then? The honest answer is no particular reason other than there being some inexplicable attraction. One quickly reads about the importance of the subject in maths and theoretical physics. I have yet to discover that, but for now I am satisfied with the curiosity it ignited in me.

Structure

For each chapter I will start by stating my thoughts and questions I had whilst reading the main text. I will indicate this by ***Pensées***. These comments might be silly and hopefully I will look back at them later and be able to answer them. When I (think I) know the answer, still, I will leave the questions here and edits by future me will be indicated in red.

I will attempt to solve all the problems. I would like to stress that my solutions will be far from polished and aim to show my thinking. Therefore, in the solutions, too, there will be modifications in red making them possibly lengthy.

With the disclaimers out of the way, let's get started.

¹For someone reading this document and spotting errors, please let me know: kianshams2001@gmail.com.

1 The modular group

Pensées

- It is explained that 'modular' refers to the moduli space of complex elliptic curves, but what is 'form' actually? I know of differential forms, bi-linear forms etc. and then how are these modular forms related to said moduli space? Hopefully the relationship with elliptic curves will be clear when reaching the corresponding chapter.
- I'm curious why $\mathbb{R}$ -linear independence is necessary in the definition of a lattice. Although they do not explicitly define this, I think $\omega_1, \omega_2 \in \mathbb{C}$ are $\mathbb{R}$ -linearly dependent if $\exists \lambda \in \mathbb{R}$ such that $\omega_1 = \lambda \omega_2$ (or vice versa). I think this is the 'full rank' part of a lattice in $\mathbb{C}$ in the definition, because else the lattice would be one-dimensional. If so, making this explicit would have been nice. **Also because any lattice $\omega_1\mathbb{Z} + \omega_2\mathbb{Z} = \omega_2\Lambda_z$ for $z \in \mathbb{C} \setminus \mathbb{R}$ by the assumption of $\mathbb{R}$ -linear independence. Wlog $z = \omega_1/\omega_2 \in \mathbb{H}$ so any homogeneous function $F : \mathcal{L} \rightarrow \mathbb{C}$ is completely determined by its values on Λ_z so induces a function $f : \mathbb{H} \rightarrow \mathbb{C}$ via $f = F(\Lambda_z)$ for $z \in \mathbb{H}$ Also, why then are we only concerned with full rank lattices?**
- Cool! Lattices in $\mathbb{C}$ are in 1-to-1 with complex elliptic curves. Curious to learn how that works later. (Actually, I also know already (merely as a trivia fact) that complex elliptic curves are Riemann surfaces. For the example of a torus I know how a lattice corresponds to it. Still, the general case will become clear later I hope.)
- They say that one is naturally led to study a function $F : \mathcal{L} \rightarrow \mathbb{C}$ satisfying $F(\lambda\Lambda) = F(\Lambda)$, $\forall \lambda \in \mathbb{C}^*$ (this is modulo homothety), but turns out to be too restrictive. Hopefully, it will be clear why that is the case. Not explained at this point. But accepting this fact, homogeneous functions of weight k are the next best alternative (?) which motivates the study of modular forms later (I secretly already know a tiny bit about it, e.g. the definition). Example of homogeneous function of weight k is the Eisenstein series, which I think is also a modular form and is relevant in physics (read: string theory) too.

1.1

- (i) " $\Leftarrow$ ": This direction we assume $\begin{pmatrix} \omega'_1 \\ \omega'_2 \end{pmatrix} = \gamma \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix}$ for some $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}_2(\mathbb{R})$. Then $\omega'_1 = a\omega_1 + b\omega_2$ and $\omega'_2 = c\omega_1 + d\omega_2$, therefore $\Lambda' \subseteq \Lambda$.² Of course γ^{-1} exists so the converse also holds, i.e. $\Lambda' = \Lambda$.

" $\Rightarrow$ ": $\Lambda' = \Lambda \Rightarrow \exists a, b, c, d \in \mathbb{Z}$ such that $\omega'_1 = a\omega_1 + b\omega_2$ and $\omega'_2 = c\omega_1 + d\omega_2$. Likewise $\exists \alpha, \beta, \delta, \theta \in \mathbb{Z}$ such that $\omega_1 = \alpha\omega'_1 + \beta\omega'_2$ and $\omega_2 = \delta\omega'_1 + \theta\omega'_2$. Plugging these relations in the first yields

$$\begin{aligned} a(\alpha\omega'_1 + \beta\omega'_2) + b(\delta\omega'_1 + \theta\omega'_2) &\stackrel{!}{=} \omega_1 \\ \Rightarrow a\beta + b\theta &= 0 \wedge a\alpha + b\delta = 1, \quad \text{likewise} \\ c(\alpha\omega'_1 + \beta\omega'_2) + d(\delta\omega'_1 + \theta\omega'_2) &\stackrel{!}{=} \omega_2 \\ \Rightarrow c\alpha + d\delta &= 0 \wedge c\beta + d\theta = 1. \end{aligned}$$

²Although the notation $\Lambda' \subseteq \Lambda$ is not introduced in the notes, I hope it is clear what I mean: all lattice points of Λ' are contained in Λ . This is different from homothety.

Compactly this is simply the requirement that

$$\underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{\gamma} \begin{pmatrix} \alpha & \beta \\ \delta & \theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (1)$$

In other words $\gamma \in \mathrm{GL}_2(\mathbb{Z})$.

- (ii) At first glance I don't understand what proposition 1.2 has to do with this part as stated in the hint... By elimination: part (i) and (ii) probably aren't useful...

Ah ok I see now, because for all $\gamma \in \mathrm{GL}_2(\mathbb{Z})$ $\det(\gamma) = \pm 1$. In order for $z' = \omega'_1/\omega'_2$ to remain in $\mathbb{H}$ we must have $\mathrm{Im}(z') > 0$. By proposition 1.2 (i) we have $\mathrm{Im}(\gamma z) = \frac{\det(\gamma)\mathrm{Im}(z)}{|j(\gamma, z)|^2} = \mathrm{Im}(z')$.

So we must have $\det(\gamma) > 0$, meaning $\det(\gamma) \stackrel{!}{=} 1$ or in other words $\gamma \in \mathrm{SL}_2(\mathbb{Z})$.

To make this proof complete really is just a copy of part (ii) and including this minor observation.

1.2

Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\gamma' = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$ be elements of $\mathrm{GL}_2(\mathbb{R})$. Then we simply work out

$$\begin{aligned} \gamma(\gamma' z) &= \gamma \left(\frac{a'z + b'}{c'z + d'} \right) = \frac{a \left(\frac{a'z + b'}{c'z + d'} \right) + b}{c \left(\frac{a'z + b'}{c'z + d'} \right) + d} \\ &= \frac{aa'z + ab' + bc'z + bd'}{ca'z + b'c + dc'z + dd'} \\ &= \frac{(aa' + bc')z + ab' + bd'}{(ca' + dc')z + b'c + dd'} =: (*). \end{aligned}$$

On the other hand,

$$\begin{aligned} (\gamma\gamma') \cdot z &= \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \right) \cdot z = \begin{pmatrix} aa' + bc' & ab' + bd' \\ a'c + c'd & b'c + dd' \end{pmatrix} \cdot z \\ &= \frac{(aa' + bc')z + ab' + bd'}{(a'c + c'd)z + b'c + dd'} = (*). \text{ qed.} \end{aligned}$$

1.3

I am confused by the two parts of this question, because a *transitive action* of a group G on a set X means that for any $x, y \in X$ there exists a $g \in G$ such that $g \cdot x = y$. However, part b of this question then asks to show that for certain γ there are fixed points. How can the action be transitive and have a fixed point?

I am confusing two things here. If there is a γ for which there is an $x \in \mathbb{H}$ such that $\gamma \cdot x = x$ then there could still be a γ' such that $\gamma' \cdot x \neq x$ of course.

a. My initial idea was to brute force: compute $\gamma \cdot z = z'$ for $z, z' \in \mathbb{H}$ and show $\mathrm{Re}(\gamma \cdot z) = \mathrm{Re}(z')$, $\mathrm{Im}(\gamma \cdot z) = \mathrm{Im}(z')$ is always solvable, but this becomes messy.

There is a much cleaner idea which I owe to David C. Ullrich from the blogpost:

<https://math.stackexchange.com/questions/2529664/how-to-show-the-action-of-sl-2-mathbb-r-on-the-complex-upper-half-plane-is-t>. Namely, for any two points z, z' on the upper half plane first map e.g. point z' to i and then i to z . Because (**CLAIM**) we can map i to any point in the upper half plane by $\mathrm{SL}_2(\mathbb{R})$ and then we simply invert this and compose the two to find a map between the two points z, z' .

Let's prove **CLAIM**. First we need a map that sends $i \mapsto \lambda i$ for $\lambda \in \mathbb{R}_{>0}$ to fix the imaginary part and then $i \mapsto i + x$ for $x \in \mathbb{R}$ to fix the real part. The group elements that do this are $\begin{pmatrix} \sqrt{\lambda} & 0 \\ 0 & 1/\sqrt{\lambda} \end{pmatrix}$

and $\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$ which are both in $\mathrm{SL}_2(\mathbb{R})$. This proves $i \mapsto x + \lambda i$ upon composition.

Let's make it explicit that we can now use this to map any two points in the upper half plane with an $\mathrm{SL}_2(\mathbb{R})$ element:

Write $z' = \alpha + \beta i$, then $i \mapsto z'$ is implemented by

$$\begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{\beta} & 0 \\ 0 & 1/\sqrt{\beta} \end{pmatrix} = \begin{pmatrix} \sqrt{\beta} & \alpha/\sqrt{\beta} \\ 0 & 1/\sqrt{\beta} \end{pmatrix} \quad (2)$$

so $z' \mapsto i$ is implemented by $\begin{pmatrix} \sqrt{\beta} & \alpha/\sqrt{\beta} \\ 0 & 1/\sqrt{\beta} \end{pmatrix}^{-1} = \begin{pmatrix} 1/\sqrt{\beta} & -\alpha/\sqrt{\beta} \\ 0 & \sqrt{\beta} \end{pmatrix}$. Likewise write $z = \epsilon + \delta i$ then $i \mapsto z$ via

$$\begin{pmatrix} \sqrt{\delta} & \epsilon/\sqrt{\delta} \\ 0 & 1/\sqrt{\delta} \end{pmatrix} \quad (3)$$

and to check we have

$$\begin{pmatrix} \sqrt{\delta} & \epsilon/\sqrt{\delta} \\ 0 & 1/\sqrt{\delta} \end{pmatrix} \begin{pmatrix} 1/\sqrt{\beta} & -\alpha/\sqrt{\beta} \\ 0 & \sqrt{\beta} \end{pmatrix} \cdot z' \quad (4)$$

$$= \begin{pmatrix} \sqrt{\frac{\delta}{\beta}} & -\alpha\sqrt{\frac{\delta}{\beta}} + \epsilon\sqrt{\frac{\beta}{\delta}} \\ 0 & \sqrt{\frac{\beta}{\delta}} \end{pmatrix} \cdot z' \quad (5)$$

$$= \frac{\sqrt{\frac{\delta}{\beta}} z' - \alpha\sqrt{\frac{\delta}{\beta}} + \epsilon\sqrt{\frac{\beta}{\delta}}}{\sqrt{\frac{\beta}{\delta}}} \quad (6)$$

$$= \frac{\delta}{\beta}(\alpha + \beta i) - \alpha\frac{\delta}{\beta} + \epsilon \quad (7)$$

$$= \epsilon + \delta i = z. \quad (8)$$

Where $\begin{pmatrix} 1/\sqrt{\beta} & -\alpha/\sqrt{\beta} \\ 0 & \sqrt{\beta} \end{pmatrix} \in \mathrm{SL}_2(\mathbb{R})$ proving transitivity of the action on $\mathbb{H}$.

b. I immediatly note that $a + d$ is the trace of the matrix, but what does this have to do with the fixed points? (**nothing immediate I think**)

Next, I thought $\gamma \cdot z \stackrel{!}{=} z$ does not yield a relation including an $a + d$ term, but $\gamma^2 \cdot z \stackrel{!}{=} \gamma \cdot z$ does. Then I collected terms and naively imposed the coefficient of z^2 terms to match, but this led to an inconsistency. That in turn made me realise we need to be careful because z^2 may not be in $\mathbb{H}$

eventhough z is (e.g. $z \in \mathbb{H}$ for $\operatorname{Re}(z) \in \mathbb{R}_{<0}$). So, I went back to $\gamma \cdot z \stackrel{!}{=} z$ to write

$$\begin{aligned}\gamma \cdot z &= \frac{az + b}{cz + d} \stackrel{!}{=} z \\ \iff cz^2 + (d - a)z - b &= 0\end{aligned}$$

Recalling secondary-school mathematics, the roots are

$$z = \frac{-(d - a) \pm \sqrt{(d - a)^2 - 4c(-b)}}{2c} \quad (9)$$

and realising that $z \in \mathbb{H}$ we must have $(d - a)^2 - 4c(-b) < 0$ and only the $+$ in $\pm$ is in the upper half plane (corresponding to there only being one fixed point). Writing out the condition

$$\begin{aligned}(d - a)^2 - 4c(-b) &= d^2 - 2ad + a^2 + 4bc \\ &= d^2 + a^2 + 2bc - 2 \quad (\text{using } ad - bc = 1) \\ &= (a + d)^2 - 2ad + 2bc - 2 \\ &= (a + d)^2 - 4 < 0 \iff |a + d| < 2.\end{aligned}$$

(N.B. This solution excludes $\gamma = \pm 1$, as stated in the problem, because then $c = 0$ which causes problems in equation 9.)

1.4

a. Write $H := \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \mid a, b \in \mathbb{R}, a^2 + b^2 = 1 \right\}$. Then clearly $H \subseteq K$, because

$$\begin{aligned}\begin{pmatrix} a & b \\ -b & a \end{pmatrix} \cdot i &= \frac{ai + b}{-bi + a} \\ &= \frac{(ai + b)(bi + a)}{a^2 + b^2} \\ &= -ab + a^2i + b^2i + ab \\ &= (a^2 + b^2)i = i.\end{aligned}$$

Conversely,

$$\begin{aligned}\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot i &= \frac{ai + b}{ci + d} \stackrel{!}{=} i \\ \iff ai + b &= -c + di \\ \iff a = d \wedge b &= -c.\end{aligned}$$

Hence, $H = K$.

b. First I was categorically confused here because the quotient $\operatorname{SL}_2(\mathbb{R})/K$ is a group, so is $\mathbb{H}$ that too? Well, of course not with regular multiplication by the observation of my solution to 1.3b (z^2 need not be in $\mathbb{H}$ when z is). So it is purely set theoretical here.

Surjectivity follows from my solution 1.3a, because for any $i \neq z \in \mathbb{H}$ we can choose a $\gamma \in \operatorname{SL}_2(\mathbb{R})/K$ similar to the construction in equation 2. For $z = i$ we simply choose the identity of course.

Recall injectivity means $\phi(a) = \phi(b) \implies a = b$ for a map ϕ and a, b in the domain of ϕ . Well, then suppose $\gamma \cdot i = \gamma' \cdot i$. This means

$$\begin{aligned} \gamma'^{-1} \cdot (\gamma \cdot i) &= (\gamma'^{-1} \circ \gamma) \cdot i && \text{(by prop 1.2.iii)} \\ &\stackrel{!}{=} i \end{aligned}$$

so in other words $\gamma'^{-1} \circ \gamma \in K$, hence $[\gamma] = [\gamma']$ in $\mathrm{SL}_2(\mathbb{R})/K$.

1.5

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